



Name:

Adm No:

Important notes / Exam rules

- Answer all the questions by choosing the correct answer/writing the correct answer. There is only one correct answer for each question.
- Each correct answer would fetch **one mark** and no partial credit.
- Tick (✓) for the correct answer.
- Notations have their usual meaning.

Q 1 Given that $y_1(t) = t$ is a solution of $t^2y'' + 2ty' - 2y = 0$. The general solution is $c_1t + c_2t^{-2}$.

Q 2 Two linearly independent solutions of $y'' - 3y' + 10y = 0$ are given by

☒ $e^{\frac{3x}{2}} \sin\left(\frac{\sqrt{31}x}{2}\right), e^{\frac{3x}{2}} \cos\left(\frac{\sqrt{31}x}{2}\right)$

☐ $e^{\frac{(3+\sqrt{31})x}{2}}, e^{\frac{(3-\sqrt{31})x}{2}}$

☐ $e^{-\frac{3x}{2}} \sin\left(\frac{\sqrt{31}x}{2}\right), e^{-\frac{3x}{2}} \cos\left(\frac{\sqrt{31}x}{2}\right)$

☐ $e^{\frac{\sqrt{31}x}{2}} \sin\left(\frac{3x}{2}\right), e^{\frac{\sqrt{31}x}{2}} \cos\left(\frac{3x}{2}\right)$

Q 3 The Wronskian of the functions x^2 and $x|x|$ for $x \in (-\infty, \infty)$ is

☒ 0

☐ x^2

☐ $2|x|$

☐ Does not exist

Q 4 The hypotheses of Picard's existence and uniqueness theorem hold to which of initial value problems on the rectangle $R : 1 < x < 3, -1 < y < 1$ with the initial condition $y(2) = 0$?

☐ $y' = e^{-x}y^{2/3}$

☒ $y' = e^xy^{3/2}$

☐ $y' = e^xy^{2/3}$

☐ $y' = e^{-x}y^{1/2}$

Q 5 The quadratic form $Q(x, y) = x^2 + y^2 + 2xy$ is reducible to the canonical form under the change of coordinates $(x, y)^T \mapsto P(x, y)^T$ where P is

☐ $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

☒ $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

☐ $\frac{1}{2} \begin{pmatrix} \sqrt{3} & 1 \\ 1 & \sqrt{3} \end{pmatrix}$

☐ $\frac{1}{2} \begin{pmatrix} \sqrt{3} & 1 \\ 1 & -\sqrt{3} \end{pmatrix}$

Q 6 An integrating factor of $(e^{x+y} + ye^y)dx + (xe^y - 5)dy = 0$ is given by

☐ e^{-x}

☐ e^x

☐ e^y

☒ e^{-y}

Q 7 The second iteration obtained by the Picard's iteration method applied to $y'' = x + y, y(0) = 1$ is

☐ $1 + x + \frac{x^2}{2} + \frac{x^3}{6}$

☐ $1 + x + \frac{x^2}{4} + \frac{x^3}{6}$

☐ $1 + x + x^2 + \frac{x^3}{3}$

☒ $1 + x + x^2 + \frac{x^3}{6}$

Q 8 The orthogonal trajectories of the family of curves $y = ce^{-x^2}$, where c is a parameter, is given by

☒ $y^2 - \ln x = k$

☐ $y^2 - x = k$

☐ $y^2 - \ln xy = k$

☐ $y - \ln x = k$

Q 9 The general solution to the Bernoulli equation $y' = 5y - 7y^2$ is given by

☐ $y = (ce^{-7x} + \frac{5}{7})^{-1}$

☒ $y = (ce^{-5x} + \frac{7}{5})^{-1}$

☐ $y = (ce^{-5x} + \frac{5}{7})^{-1}$

☐ $y = (ce^{-7x} + \frac{7}{5})^{-1}$

Q 10 The differential equation $M(x, y)dx + N(x, y)dy = 0$ is exact for $M(x, y) = ye^{xy} \sin(e^{xy}) + ye^{\sin x} \cos x$ and $N(x, y) =$

☐ $x \sin(e^{xy})e^{xy} + e^{\cos x}$

☒ $x \sin(e^{xy})e^{xy} + e^{\sin x}$

☐ $y \sin(e^{xy})e^{xy} + e^{\sin x}$

☐ $x \cos(e^{xy})e^{xy} + e^{\sin x}$

Name:

Adm No:

Important notes / Exam rules

- Answer all the questions by choosing the correct answer/writing the correct answer. There is only one correct answer for each question.
- Each correct answer would fetch **one mark** and no partial credit.
- Tick (✓) for the correct answer.
- Notations have their usual meaning.

Q 1 The orthogonal trajectories of the family of curves $y = ce^{-x^2}$, where c is a parameter, is given by

☐ $y^2 - \ln xy = k$
☒ $y^2 - \ln x = k$
☐ $y - \ln x = k$
☐ $y^2 - x = k$

Q 2 The second iteration obtained by the Picard's iteration method applied to $y'' = x + y, y(0) = 1$ is

☒ $1 + x + x^2 + \frac{x^3}{6}$
☐ $1 + x + x^2 + \frac{x^3}{3}$
☐ $1 + x + \frac{x^2}{2} + \frac{x^3}{6}$
☐ $1 + x + \frac{x^2}{4} + \frac{x^3}{6}$

Q 3 The quadratic form $Q(x, y) = x^2 + y^2 + 2xy$ is reducible to the canonical form under the change of coordinates $(x, y)^T \mapsto P(x, y)^T$ where P is

☐ $\frac{1}{2} \begin{pmatrix} \sqrt{3} & 1 \\ 1 & -\sqrt{3} \end{pmatrix}$
☐ $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$
☒ $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
☐ $\frac{1}{2} \begin{pmatrix} \sqrt{3} & 1 \\ 1 & \sqrt{3} \end{pmatrix}$

Q 4 The general solution to the Bernoulli equation $y' = 5y - 7y^2$ is given by

☐ $y = (ce^{-7x} + \frac{5}{7})^{-1}$
☐ $y = (ce^{-7x} + \frac{7}{5})^{-1}$
☒ $y = (ce^{-5x} + \frac{7}{5})^{-1}$
☐ $y = (ce^{-5x} + \frac{5}{7})^{-1}$

Q 5 The differential equation $M(x, y)dx + N(x, y)dy = 0$ is exact for $M(x, y) = ye^{xy} \sin(e^{xy}) + ye^{\sin x} \cos x$ and $N(x, y) =$

☒ $x \sin(e^{xy})e^{xy} + e^{\sin x}$
☐ $x \cos(e^{xy})e^{xy} + e^{\sin x}$
☐ $x \sin(e^{xy})e^{xy} + e^{\cos x}$
☐ $y \sin(e^{xy})e^{xy} + e^{\sin x}$

Q 6 Two linearly independent solutions of $y'' - 3y' + 10y = 0$ are given by

☐ $e^{\frac{\sqrt{31}x}{2}} \sin\left(\frac{3x}{2}\right), e^{\frac{\sqrt{31}x}{2}} \cos\left(\frac{3x}{2}\right)$
☒ $e^{\frac{3x}{2}} \sin\left(\frac{\sqrt{31}x}{2}\right), e^{\frac{3x}{2}} \cos\left(\frac{\sqrt{31}x}{2}\right)$
☐ $e^{\frac{-3x}{2}} \sin\left(\frac{\sqrt{31}x}{2}\right), e^{\frac{-3x}{2}} \cos\left(\frac{\sqrt{31}x}{2}\right)$
☐ $e^{\frac{(3+\sqrt{31})x}{2}}, e^{\frac{(3-\sqrt{31})x}{2}}$

Q 7 The Wronskian of the functions x^2 and $x|x|$ for $x \in (-\infty, \infty)$ is

☒ 0
 ☐ Does not exist
 ☐ $2|x|$
☐ x^2

Q 8 An integrating factor of $(e^{x+y} + ye^y)dx + (xe^y - 5)dy = 0$ is given by

☐ e^{-x}
☐ e^x
☒ e^{-y}
☐ e^y

Q 9 Given that $y_1(t) = t$ is a solution of $t^2y'' + 2ty' - 2y = 0$. The general solution is $c_1t + c_2t^{-2}$.

Q 10 The hypotheses of Picard's existence and uniqueness theorem hold to which of initial value problems on the rectangle $R: 1 < x < 3, -1 < y < 1$ with the initial condition $y(2) = 0$?

☐ $y' = e^{-x}y^{1/2}$
☒ $y' = e^xy^{3/2}$
☐ $y' = e^xy^{2/3}$
☐ $y' = e^{-x}y^{2/3}$



Name:

Adm No:

Important notes / Exam rules

- Answer all the questions by choosing the correct answer/writing the correct answer. There is only one correct answer for each question.
- Each correct answer would fetch **one mark** and no partial credit.
- Tick (✓) for the correct answer.
- Notations have their usual meaning.

Q 1 The general solution to the Bernoulli equation $y' = 5y - 7y^2$ is given by

☐ $y = (ce^{-5x} + \frac{5}{7})^{-1}$
☒ $y = (ce^{-5x} + \frac{7}{5})^{-1}$
☐ $y = (ce^{-7x} + \frac{5}{7})^{-1}$
☐ $y = (ce^{-7x} + \frac{7}{5})^{-1}$

Q 2 An integrating factor of $(e^{x+y} + ye^y)dx + (xe^y - 5)dy = 0$ is given by

☐ e^y
☐ e^x
☒ e^{-y}
☐ e^{-x}

Q 3 The orthogonal trajectories of the family of curves $y = ce^{-x^2}$, where c is a parameter, is given by

☒ $y^2 - \ln x = k$
☐ $y - \ln x = k$
☐ $y^2 - x = k$
☐ $y^2 - \ln xy = k$

Q 4 The Wronskian of the functions x^2 and $x|x|$ for $x \in (-\infty, \infty)$ is

☒ 0
 ☐ Does not exist
☐ $2|x|$
☐ x^2

Q 5 The hypotheses of Picard's existence and uniqueness theorem hold to which of initial value problems on the rectangle $R: 1 < x < 3, -1 < y < 1$ with the initial condition $y(2) = 0$?

☒ $y' = e^x y^{3/2}$
☐ $y' = e^x y^{2/3}$
☐ $y' = e^{-x} y^{2/3}$
☐ $y' = e^{-x} y^{1/2}$

Q 6 Given that $y_1(t) = t$ is a solution of $t^2 y'' + 2ty' - 2y = 0$. The general solution is $c_1 t + c_2 t^{-2}$.

Q 7 The quadratic form $Q(x, y) = x^2 + y^2 + 2xy$ is reducible to the canonical form under the change of coordinates $(x, y)^T \mapsto P(x, y)^T$ where P is

☒ $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
☐ $\frac{1}{2} \begin{pmatrix} \sqrt{3} & 1 \\ 1 & \sqrt{3} \end{pmatrix}$
☐ $\frac{1}{2} \begin{pmatrix} \sqrt{3} & 1 \\ 1 & -\sqrt{3} \end{pmatrix}$
☐ $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

Q 8 Two linearly independent solutions of $y'' - 3y' + 10y = 0$ are given by

☐ $e^{\frac{\sqrt{31}x}{2}} \sin\left(\frac{3x}{2}\right), e^{\frac{\sqrt{31}x}{2}} \cos\left(\frac{3x}{2}\right)$
☒ $e^{\frac{3x}{2}} \sin\left(\frac{\sqrt{31}x}{2}\right), e^{\frac{3x}{2}} \cos\left(\frac{\sqrt{31}x}{2}\right)$
☐ $e^{\frac{-3x}{2}} \sin\left(\frac{\sqrt{31}x}{2}\right), e^{\frac{-3x}{2}} \cos\left(\frac{\sqrt{31}x}{2}\right)$
☐ $e^{\frac{(3+\sqrt{31})x}{2}}, e^{\frac{(3-\sqrt{31})x}{2}}$

Q 9 The differential equation $M(x, y)dx + N(x, y)dy = 0$ is exact for $M(x, y) = ye^{xy} \sin(e^{xy}) + ye^{\sin x} \cos x$ and $N(x, y) =$

☐ $x \sin(e^{xy})e^{xy} + e^{\cos x}$
☒ $x \sin(e^{xy})e^{xy} + e^{\sin x}$
☐ $y \sin(e^{xy})e^{xy} + e^{\sin x}$
☐ $x \cos(e^{xy})e^{xy} + e^{\sin x}$

Q 10 The second iteration obtained by the Picard's iteration method applied to $y'' = x + y, y(0) = 1$ is

☐ $1 + x + \frac{x^2}{4} + \frac{x^3}{6}$
☐ $1 + x + x^2 + \frac{x^3}{3}$
☐ $1 + x + \frac{x^2}{2} + \frac{x^3}{6}$
☒ $1 + x + x^2 + \frac{x^3}{6}$

Name:

Adm No:

Important notes / Exam rules

- Answer all the questions by choosing the correct answer/writing the correct answer. There is only one correct answer for each question.
- Each correct answer would fetch **one mark** and no partial credit.
- Tick (✓) for the correct answer.
- Notations have their usual meaning.

Q 1 Two linearly independent solutions of $y'' - 3y' + 10y = 0$ are given by

☐ $e^{-\frac{3x}{2}} \sin\left(\frac{\sqrt{31}x}{2}\right), e^{-\frac{3x}{2}} \cos\left(\frac{\sqrt{31}x}{2}\right)$

☒ $e^{\frac{3x}{2}} \sin\left(\frac{\sqrt{31}x}{2}\right), e^{\frac{3x}{2}} \cos\left(\frac{\sqrt{31}x}{2}\right)$

☐ $e^{\frac{(3+\sqrt{31})x}{2}}, e^{\frac{(3-\sqrt{31})x}{2}}$

☐ $e^{\frac{\sqrt{31}x}{2}} \sin\left(\frac{3x}{2}\right), e^{\frac{\sqrt{31}x}{2}} \cos\left(\frac{3x}{2}\right)$

Q 2 The hypotheses of Picard's existence and uniqueness theorem hold to which of initial value problems on the rectangle $R: 1 < x < 3, -1 < y < 1$ with the initial condition $y(2) = 0$?

☐ $y' = e^{-x}y^{2/3}$

☐ $y' = e^x y^{2/3}$

☐ $y' = e^{-x}y^{1/2}$

☒ $y' = e^x y^{3/2}$

Q 3 The quadratic form $Q(x, y) = x^2 + y^2 + 2xy$ is reducible to the canonical form under the change of coordinates $(x, y)^T \mapsto P(x, y)^T$ where P is

☐ $\frac{1}{2} \begin{pmatrix} \sqrt{3} & 1 \\ 1 & \sqrt{3} \end{pmatrix}$

☒ $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

☐ $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

☐ $\frac{1}{2} \begin{pmatrix} \sqrt{3} & 1 \\ 1 & -\sqrt{3} \end{pmatrix}$

Q 4 Given that $y_1(t) = t$ is a solution of $t^2 y'' + 2ty' - 2y = 0$. The general solution is $c_1 t + c_2 t^{-2}$.

Q 5 An integrating factor of $(e^{x+y} + ye^y)dx + (xe^y - 5)dy = 0$ is given by

☐ e^x

☐ e^y

☒ e^{-y}

☐ e^{-x}

Q 6 The second iteration obtained by the Picard's iteration method applied to $y'' = x + y, y(0) = 1$ is

☒ $1 + x + x^2 + \frac{x^3}{6}$

☐ $1 + x + \frac{x^2}{2} + \frac{x^3}{6}$

☐ $1 + x + x^2 + \frac{x^3}{3}$

☐ $1 + x + \frac{x^2}{4} + \frac{x^3}{6}$

Q 7 The differential equation $M(x, y)dx + N(x, y)dy = 0$ is exact for $M(x, y) = ye^{xy} \sin(e^{xy}) + ye^{\sin x} \cos x$ and $N(x, y) =$

☒ $x \sin(e^{xy})e^{xy} + e^{\sin x}$

☐ $x \cos(e^{xy})e^{xy} + e^{\sin x}$

☐ $y \sin(e^{xy})e^{xy} + e^{\sin x}$

☐ $x \sin(e^{xy})e^{xy} + e^{\cos x}$

Q 8 The Wronskian of the functions x^2 and $x|x|$ for $x \in (-\infty, \infty)$ is

☒ 0

☐ x^2

☐ Does not exist

☐ $2|x|$

Q 9 The general solution to the Bernoulli equation $y' = 5y - 7y^2$ is given by

☐ $y = (ce^{-7x} + \frac{7}{5})^{-1}$

☒ $y = (ce^{-5x} + \frac{7}{5})^{-1}$

☐ $y = (ce^{-7x} + \frac{5}{7})^{-1}$

☐ $y = (ce^{-5x} + \frac{5}{7})^{-1}$

Q 10 The orthogonal trajectories of the family of curves $y = ce^{-x^2}$, where c is a parameter, is given by

☐ $y^2 - \ln xy = k$

☐ $y - \ln x = k$

☐ $y^2 - x = k$

☒ $y^2 - \ln x = k$